[INCH-POUND]
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

GRINDING MACHINES, SURFACE, ROTARY TABLE, VERTICAL SPINDLE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers vertical spindle, rotary table, surface grinding machines. The machines covered by this CID are intended for use in shops and general production plants, for rough and finish grinding with rapid metal removal rates, and to produce flat surfaces on a wide variety of material.

2. CLASSIFICATION. The grinding machines shall be of the following types, styles, and sizes. The type, style, and size to be furnished shall be as specified (see 7.2(b)).

Type: Type I - Manually operated
Type II - Automatic cycle

Style: Style A - Single spindle, single transitional table
Style B - Single spindle, dual transitional table
Style C - Single table, single transitional spindle
Style D - Single table, dual spindle

Size: Sizes are shown in tables I–IV.

Beneficial comments, recommendations, additions, deletions, and clarifications which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3415

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TABLE I. Style A machine sizes.

Size	Table diameter, inches	Size	Table diameter, inches
010	10	060	60
012	12	066	66
014	14	072	72
016	16	084	84
020	20	096	96
026	26	108	108
030	30	120	120
036	36	144	144
042	42	160	160
048	48	168	168
054	54		

TABLE II. Style B machine sizes.

Size	Table diameter, inches
020	20
030	30
048	48
054	54

TABLE III. Style C machine sizes.

Size	Table diameter, inches
006	6
008	8
012	12
016	16

TABLE IV. Style D machine sizes.

Size	Effective grinding width[1] of table, inches
006	6
012	12

[1] Width = (outside diameter – inside diameter) ÷ 2

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3. SALIENT CHARACTERISTICS.

3.1 General requirements. The grinding machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2 Components. The grinding machine shall include, as a minimum, the following components:

3.2.1 Base. Unless otherwise specified (see 7.2(c)), the base shall be a casting, fabrication, or component assembly of functional design to support the other components and provide for minimal distortion from operational loads, stress, vibration, and temperature gradients. A means shall be provided for leveling and anchoring the machine to the floor. A means shall be provided for collection and discharge of coolant and grinding swarf from the base. A means shall be provided for access to serviceable components housed within the base.

3.2.2 Rotary table. The rotary table shall be a circular table oriented in the horizontal plane. The rotary tables on style A and B machines shall be mounted on a carriage suitable for transit along the base machineways. The distance of rotary table transit shall be from full grind position to full disengagement or loading position. The rotary tables on style C and D machines shall be mounted directly to the base. When specified (see 7.2(d)), a carriage suitable for vertical or angular adjustment shall be provided. Independent drive units and controls shall be provided for table rotation and, when applicable, for rotary table transition. When specified (see 7.2(e)), an extra table shall be provided with a means for attachment of workpiece holding fixtures. Unless otherwise specified (see 7.2(f)), the grinding machine shall have a minimum workpiece weight capacity of 250 pounds per square foot of the rotary table work surface area.

3.2.3 Magnetic chuck. The magnetic chuck shall be the same diameter as and be securely mounted on the rotary table. When mounted, the magnetic chuck shall be properly aligned to ensure flatness requirements of the finished work. Unless otherwise specified (see 7.2(g)), the chuck shall be an electromagnet type made of steel and machined and fitted with coils that are adequately insulated and sealed against coolant ingress or condensate formation. The magnetic chuck shall securely hold the workpiece to the face of the chuck during maximum metal removal conditions using the manufacturer’s recommended loading and blocking procedures.

3.2.4 Wheelhead column. Unless otherwise specified (see 7.2(h)), the wheelhead column shall be a singular casting or fabricated weldment designed to minimize distortion and deflection. The mating surface of the column, or the support flange, shall provide for alignment and shall be bolted to the base, forming single unit rigidity. The column shall accommodate a vertically traveling grinding wheelhead assembly, speed and feed components, driving mechanisms, and, when applicable, a wheel dresser and controls. Access shall be provided to serviceable components housed within the column.

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3.2.5 Wheelhead assembly. The wheelhead assembly shall include a wheelhead housing, grinding wheel or segmented grinding wheel assembly, grinding wheel spindle, spindle drive unit, vertical positioning drive unit, and downfeed drive unit. The spindle drive shall provide peripheral grinding wheel speeds within the range of 3,400 to 6,000 surface feet per minute. The spindle shall be mounted on antifriction bearings designed to minimize axial and radial play and shall be dynamically balanced. The spindle bearings shall be continuously or permanently lubricated. Unless otherwise specified (see 7.2(i)), rapid positioning of the wheelhead assembly shall be in accordance with the manufacturer's standard practice. On type I grinding machines, both course and fine downfeed shall be provided. When specified (see 7.2(j)), the grinding machine shall be provided with a power actuated tilting wheelhead. When specified (see 7.2(k)), the spindle shall be provided with dynamic braking to ensure rapid, smooth stopping of the grinding wheel.

3.2.6 Machineways. Machineways shall be provided to support transitional tables, carriages, and drives. Machineways shall be protected regardless of grinding machine position to exclude chips, dust, coolant, and other contaminants or hazards that may cause damage or result in inaccurate performance. Machineways shall be substantial enough to prevent the grinding machine from deviating beyond alignment limits during operation at maximum capacity.

3.2.7 Grinding wheel. Style A, B, and C grinding machines shall have a grinding wheel diameter equal to or greater than half of the rotary table diameter. Style D grinding machines shall have a grinding wheel diameter equal to or greater than the effective width of the rotary table. Unless otherwise specified (see 7.2(l)), a solid cylindrical or segmented grinding wheel shall be provided in accordance with the manufacturer's standard issue for the grinding machine provided. Grinding wheels shall meet the requirements of American National Standards Institute (ANSI) B7.1.

3.2.8 Wheel dresser. Unless otherwise specified (see 7.2(m)), a wheel dresser capable of dressing the entire working face of both solid cylindrical and segmented grinding wheels shall be provided.

3.2.9 Controls.

3.2.9.1 Manual control (type I). Unless otherwise specified (see 7.2(n)), the grinding machine shall be provided with an oil-tight overhead pendent or push-button station located convenient to the normal operators position. As a minimum, the following manual controls shall be provided:

a. Grinding wheel start/stop (rotation).
b. Table or carriage in/out/jog (hand-feed is permissible for machines without automatic cycle).
c. Chuck start/stop (rotation).
d. Wheelhead up/down/jog (hand-feed is permissible for machines without automatic cycle).
e. Coolant start/stop.
f. Emergency stop (entire machine).
g. Chuck on/off (magnetic).
h. When applicable, automatic cycle start/stop.

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3.2.9.2 Automatic control (type II). The grinding machine shall be provided with automatic cycle operation. After predetermined machine and workpiece parameter input, loading of the workpiece, and activation of the magnetic chuck, the machine shall complete an automatic grind cycle by command of a single control. Unless otherwise specified (see 7.2(o)),, the automatic cycle shall include, as a minimum, the following functions:

a. Moving chuck to grinding position.
b. Starting chuck rotation.
c. Starting grinding wheel rotation.
d. Engaging coolant pump.
e. Engaging power downfeed at preset rate until stopped by automatic gauging or preset limit, including control of head tilt feature when applicable.
f. Accomplishing preset timed sparkout.
g. Retracting wheelhead to preset location.
h. Stopping grinding wheel rotation and coolant pump.
i. Retracting chuck to load position (when applicable), stopping chuck rotation, and demagnetizing chuck.
j. Operator override, with the automatic cycle capable of being interrupted and reentered during the run cycle.

3.2.9.3 Automatic sizing. When specified (see 7.2(p)), an automatic sizing system shall be provided. The sizing system shall give continuous measurement of the workpiece during grinding and a visual indication that the workpiece is approaching size. When the workpiece reaches a preset dimension, feed shall stop, the machine shall sparkout, and the grinding wheel shall retract.

3.2.9.4 Rotary dials or handwheels. Unless otherwise specified (see 7.2(q)), rotary dials or handwheels that are provided to position the machine axis for grinding operations shall be graduated in increments not larger than 0.0001 inch and shall be easily visible from the operators position. The feed indicators shall be capable of individually indexing to zero at any point and contiguously calibrated over the full range. Dial and wheel markings shall be permanently and legibly engraved or etched on a nonglare surface.

3.2.9.5 Digital readout. When specified (see 7.2(r)), a digital readout shall be provided to reference the wheelhead position. The digital readout system shall consist of an electronic display, related components, and any equipment necessary to provide a readily visible digital readout over the full working range. The digital readout shall be capable of indicating in both US and SI systems of measurement. Unless otherwise specified (see 7.2(s)), the digital readout system shall meet the requirements shown in table V.

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TABLE V. Digital readout system requirements.

Parameter	Requirement
Resolution	0.00001 inch
Accuracy	±0.00005 inch
Repeatability	±0.00005 inch

3.2.10 Lubrication system. A lubrication system shall be provided to ensure lubrication of all moving parts. When specified (see 7.2(t)), the lubrication system shall be completely automatic and shall be provided with interlocks to prevent operation in case of pressure loss. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing.

3.2.11 Coolant system. The coolant system shall be of sufficient capacity to provide full flow rate during the entire grinding process. Unless otherwise specified (see 7.2(u)), the coolant system shall be the flood type. The system shall be equipped to allow the operator to direct and control the coolant flow to the work and to flush the base. When specified (see 7.2(v)), fittings and valves for directing coolant to the inside of the grinding wheel shall be provided.

3.2.12 Hydraulic system. When specified (see 7.2(w)), a hydraulic system shall be provided. When a hydraulic system is provided, the following conditions shall apply. Overpressure protection shall be provided to prevent damage to components. A filter system shall ensure delivery of clean fluid to the hydraulic system. The hydraulic system reservoir shall have a fluid level indicator. The hydraulic fluid temperature shall be maintained within the proper operating range during grinding machine operation. A means shall be provided to drain and clean the hydraulic reservoir.

3.2.13 Electrical system. Unless otherwise specified (see 7.2(x)), the electrical system shall conform to the ANSI/National Fire Protection Association (NFPA) 79 or International Electrotechnical Commission (IEC) 204-1. Unless otherwise specified (see 7.2(y)), the machine shall operate from a dual 230/460-volt, 3-phase, 60-hertz circuit and shall be initially wired for 230-volt operation. The control system shall operate on alternating current, 120-volts or less, single-phase, obtained from a transformer with an isolated secondary winding.

3.2.14 Electric motors. The grinding wheel, coolant pump, and dust extractor drive motor(s) shall have continuously lubricated bearings or sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG-1. Unless otherwise specified (see 7.2(z)), the motors shall have open drip-proof enclosures.

3.2.15 Optional accessories. Optional accessories shall be furnished as specified (see 7.2(aa)).

3.3 Performance. Unless otherwise specified (see 7.2(bb)), the grinding machine shall meet the alignment and performance requirements shown in tables VI and VII, respectively, when tested in accordance with sections 5.3 through 5.3.5.

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TABLE VI. Alignment requirements.

Parameter	Tolerance
Worktable or chuck flat in all directions	0.0001 inch/foot
Axial runout of worktable or chuck: sizes up to 14 sizes over 14	0.0002 inch TIR[1] 0.0006 inch TIR
Surface of rotary table parallel with its traverse (style A and B) or parallel with wheel traverse (style C and D)	0.002 inch TIR
Taper of wheel spindle true running	0.0002 inch TIR
Axial runout of wheel spindle	0.0002 inch TIR
Wheel spindle square with rotary table (measured at two points with 90 degrees rotation)	0.0015 inch/foot elevation
Vertical travel of wheelhead square with table	0.0015 inch/foot travel

[1] TIR = Total indicator reading.

TABLE VII. Performance requirements.

Parameter	Styles A, B, and D	Style C
Rapid metal removal	$(0.008 \times A^{1})$ cubic-inches/minute	
Parallel surfaces	0.0002 inch/foot	0.0005 inch/foot
Finished dimensions	0.0002 inch/foot	0.0005 inch/foot
Finished surface, total roughness	8 microinch Rt[2]	16 microinch Rt

[1] A = Numeric value for working area (in square inches) of rotary table top.
[2] Rt = Roughness total as defined in American Society of Mechanical Engineers (ASME) B46.1.

3.4 Work zone. The grinding machine work zone is defined by the working area of the rotary table and the height (clearance) of a new grinding wheel above the table. The minimum work zone height shall be as specified (see 7.2(cc))

3.5 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(dd)).

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3.6 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI B11.9 or an equivalent standard from the country of manufacture. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910. If a conflict arises between ANSI B11.9 or its equivalent and the OSHA standards, the OSHA standards shall apply.

3.7 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(ee)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(ff)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.8 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(gg)), the nameplate shall contain the following information:

a. Nomenclature.
b. Manufacturer's name.
c. Serial number.
d. Machine model designation.
e. Power input (volts, total amperes, phase, frequency).
f. Short-circuit/over-current rating.
g. Contract number or order number.
h. National stock number or plant equipment code.
i. Date of manufacture.

3.9 Lubrication plate or chart. When specified (see 7.2(hh)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

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5. QUALITY ASSURANCE PROVISIONS.

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The grinding machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(ii)), the grinding machine shall be tested in accordance with sections 5.3.1 through 5.3.5.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(jj)), the manufacturer shall be responsible for supplying all tooling and material(s) and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(kk)). Unless otherwise specified (see 7.2(ll)), the machine shall be tested in accordance with environmental conditions of 40 to 104 degrees Fahrenheit and 20 to 95 percent humidity.

5.3.2 Operational test. The machine and its supplied systems shall be operated in accordance with the manufacturer's standard operating procedures for warm-up and run-off checks. The machine shall be operated at no load for not less than 6 hours. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Alignment tests. Alignment shall be verified before and after the performance tests. Changes in alignment shall be cause for rejection.

5.3.4 Performance tests. Unless otherwise specified (see 7.2(mm)), the tests listed in sections 5.3.4.2 through 5.3.4.3 shall be conducted in accordance with the manufacturer's recommendations and procedures.

5.3.4.1 Rapid metal removal. Unless otherwise specified (see 7.2(nn)), the grinding machine shall be capable of rapid metal removal. Rapid metal removal shall be tested by rough grinding the top surface of a plate(s) of cast iron, and removing the volume of metal per minute required in table VII of section 3.3.

5.3.4.2 Surface grinding. Eight equal-sized test blocks of cast iron shall be surface ground simultaneously on each machine table. Four blocks shall be positioned equidistant apart 3 to 4 inches from the outer periphery of the table. Four blocks shall be positioned equidistant apart 3 to 6 inches from the center or along the inner periphery of the table. The size of the test blocks shall vary in accordance with the size of the machine, so that at least 20 percent of the table area is covered by the test blocks. After finish grinding the first side, the blocks shall be inverted at the same location on the table for the finish grind of the second side. The test results shall be equal to or better than the performance required in table VII of section 3.3.

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5.3.4.3 Surface finish. SAE O1 tool steel hardened to 59-60 on the Rockwell hardness C scale, shall be substituted for the cast iron in two of the test blocks during the surface grinding for testing surface finish requirements. The grinding machine shall produce surfaces with a total roughness no greater than shown in table VII on the test blocks composed of SAE O1 tool steel.

5.3.5 Optional tests. Optional tests shall be conducted as specified (see 7.2(oo)).

5.4 Acceptance. Unless otherwise specified (see 7.2(pp)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the performance tests shall be compared with the performance requirements shown in tables VI and VII. Failure of the machine to meet the performance requirements shall be cause for rejection.

6. PACKAGING. Preservation, packing, and marking shall be as specified (7.2(qq)).

7. NOTES

7.1 Referenced documents.

7.1.1 Government documents. Copies of Federal documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.1.2 Industry standards. Copies of industry standards referenced in this CID may be obtained from the following addresses:

American National Standards Institute (ANSI)

ANSI B11.9 Safety Requirements for the Construction, Care, and Use of Grinding Machines

ANSI B7.1 Safety Requirements for the Use, Care, and Protection of Abrasive Wheels

Application for copies should be sent to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

American Society of Mechanical Engineers (ASME)

ASME B46.1 Surface Texture (Surface Roughness, Waviness, and Lay)

Application for copies should be sent to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.

International Electrotechnical Commission (IEC)

IEC 204-1 Electrical equipment of industrial machines - Part 1: General requirements

Applications for copies should be sent to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

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Occupational Safety and Health Association (OSHA)

OSHA 1910 Occupational Safety and Health Standards for General Industry (CFR 29 PART 1910)

Application for copies should be sent to the U. S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

National Electrical Manufacturer's Association (NEMA)

NEMA MG - 1 NEMA standards publication - Motors and Generators

Applications for copies should be sent to the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Roslyn, VA 22209.

National Fire Protection Association (NFPA)

ANSI/NFPA 79 Electrical Standards for Industrial Machinery

Applications for copies should be sent to the National Fire Protection Association, One Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Type, style, and size of grinding machine required (see 2).
c. Base requirements, if different (see 3.2.1).
d. Table carriage requirements for vertical/angular adjustment, if required (see 3.2.2).
e. Extra table, if required (see 3.2.2).
f. Weight capacity, if different (see 3.2.2).
g. Chuck, if different (see 3.2.3).
h. Wheelhead column, if different (see 3.2.4).
i. Rapid positioning, if different (see 3.2.5).
j. Power actuated tilt option, if required (see 3.2.5).
k. Dynamic spindle braking, if required (see 3.2.5).
l. Grinding wheel type, if different (see 3.2.7).
m. Wheel dresser, if different (see 3.2.8).
n. Manual controls, if different (see 3.2.9.1).
o. Automatic cycle, if different (see 3.2.9.2).
p. Automatic sizing system, if required (see 3.2.9.3).
q. Indicator/dial graduations, if different (see 3.2.9.4).
r. Digital readout system, if required (see 3.2.9.5).
s. Digital readout requirements, if different (see 3.2.9.5).
t. Fully automatic lubricating system, if required (see 3.2.10).
u. Coolant system, if different (see 3.2.11).
v. Coolant delivery internal to grinding wheel, if required (see 3.2.11).
w. Hydraulic system, if required (see 3.2.12).
x. Electrical system conformance, if different (see 3.2.13).
y. Voltage, if different (see 3.2.13).

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z. Motor enclosure, if different (see 3.2.14).
aa. Optional accessories, as specified (see 3.2.15).
bb. Performance, if different (see 3.3).
cc. Minimum work zone height, as specified (see 3.4).
dd. Machine dimensions and weight, as specified (see 3.5).
ee. System of units required (see 3.7).
ff. Dual system of units, if required (see 3.7).
gg. Nameplate, if different (see 3.8).
hh. Lubrication plate or chart, if required (see 3.9).
ii. Testing, if different (see 5.3).
jj. Test responsibility, if different (see 5.3.1).
kk. Certification of calibration, as specified (see 5.3.1).
ll. Test environmental conditions, if different (see 5.3.1).
mm. Performance tests, if different (see 5.3.4).
nn. Rapid metal removal, if different (see 5.3.4.1).
oo. Optional test, if required (see 5.3.5).
pp. Acceptance, if different (see 5.4).
qq. Packaging requirements, as specified (see 6).

7.3 Part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.

A-A-59200 -XXXXX

Size: (See tables I through IV for list of sizes)

Style: A - Style A, Single spindle, single transitional table
B - Style B, Single spindle, dual transitional table
C - Style C, Single table, single transitional spindle
D - Style D, Single table, dual spindle

Type: 1 - Type I, Manually operated
2 - Type II, Automatic cycle

CID number

MILITARY INTERESTS:

Custodians
Air Force - 99
Army - AL
Navy - SH

Reviewers
Air Force - 84
Navy - AS, MC, OS

PREPARING ACTIVITY:
DLA - GS

(Project 3415-0215)

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